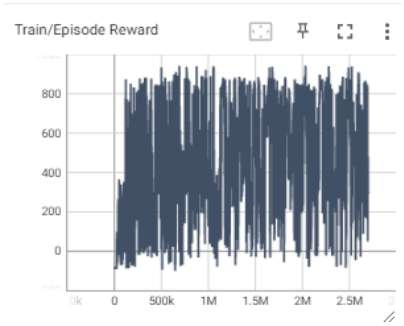
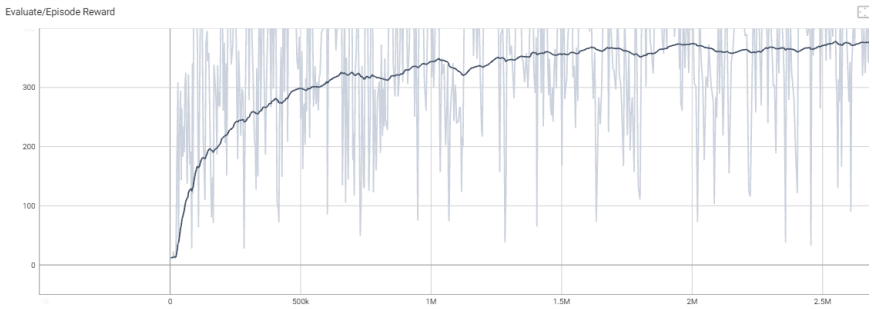
- Experimental Results (30%)

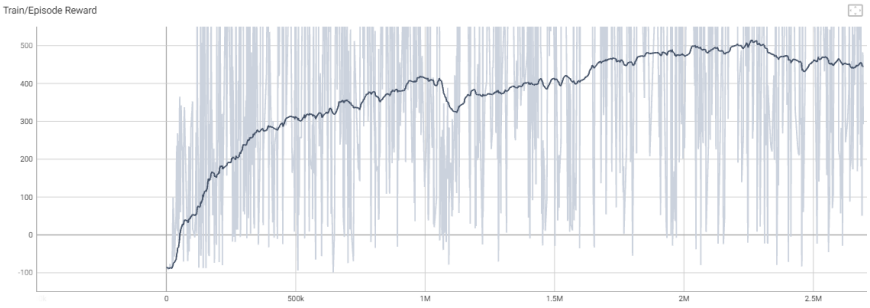
  Based on Gaussian Noise

  1. Curve



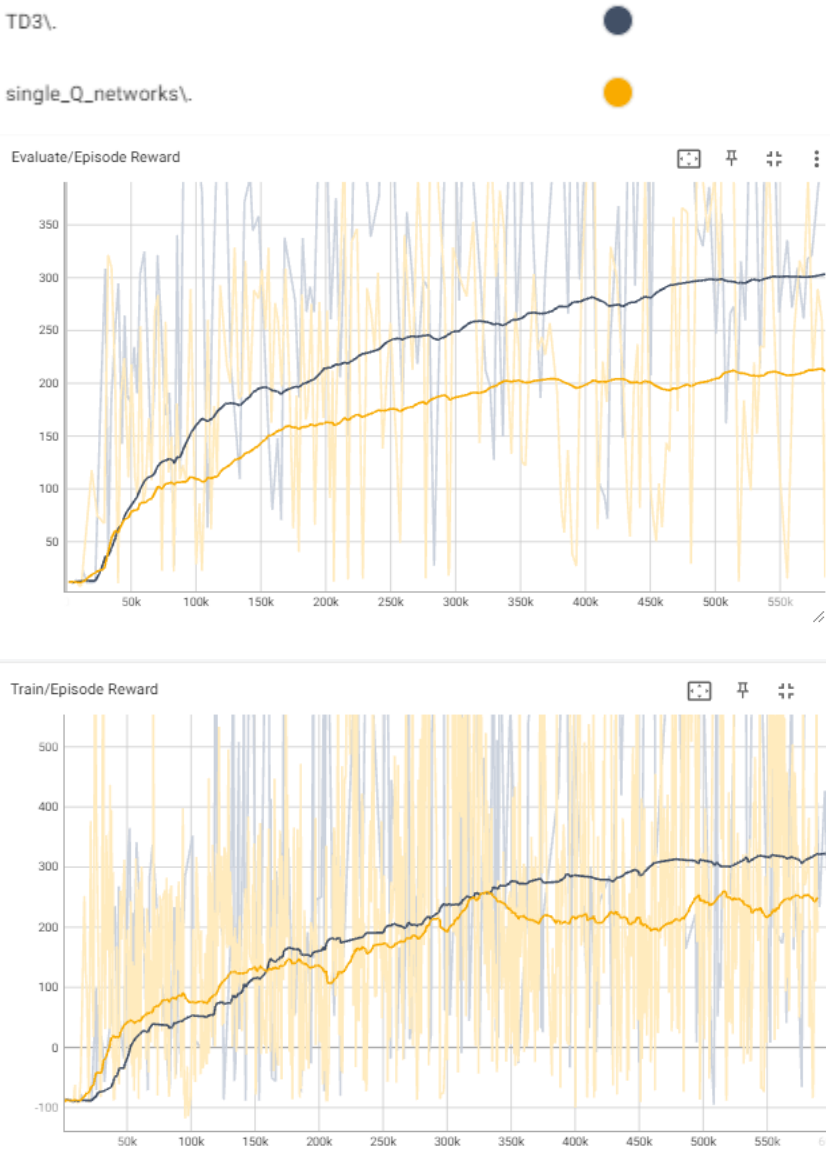




  2. testing results

```
Evaluating...
Episode: 1      Length: 999     Total reward: 820.27
Episode: 2      Length: 258     Total reward: 260.25
Episode: 3      Length: 297     Total reward: 336.54
Episode: 4      Length: 809     Total reward: 846.66
Episode: 5      Length: 281     Total reward: 307.33
Episode: 6      Length: 927     Total reward: 847.27
Episode: 7      Length: 587     Total reward: 683.34
Episode: 8      Length: 311     Total reward: 385.76
Episode: 9      Length: 999     Total reward: 801.10
Episode: 10     Length: 999     Total reward: 807.89
average score: 609.6410190635555
```

- Experimental Results and Discussion of bonus parts (Impact of Twin Q-Networks, Target Policy Smoothing, Delayed Policy Update Mechanism, Action Noise Injection) (bonus)(30%)

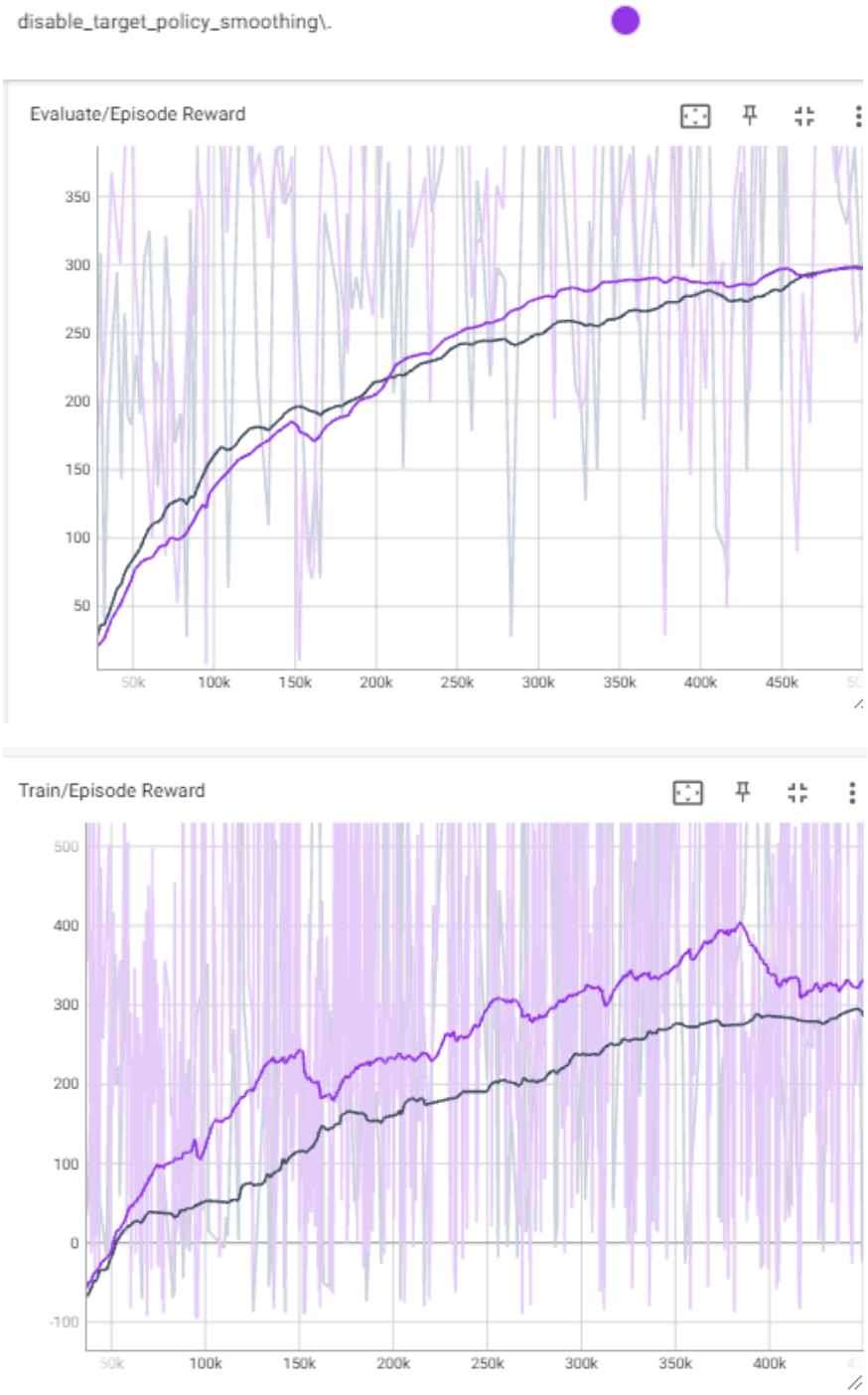
1. Screenshot of Tensorboard training curve and compare the performance of using twin Q-networks and single Q-networks in TD3, and explain(5%).




從 Evaluation 的圖表可以看見 Single Q-networks 得到的 reward 比 twin Q-networks 還要低，推測 Single Q-networks 可能更容易在估計 Q-value 中出現錯誤，導致 exploration 的 policy 不理想。
另外，Single Q-networks 的 TD3 的 reward curve 較早趨於平緩，可見 Single Q-networks 的 TD3 會比較早進入收斂。

2. Screenshot of Tensorboard training curve and compare the impact of enabling and disabling target policy smoothing in TD3, and explain(5%).

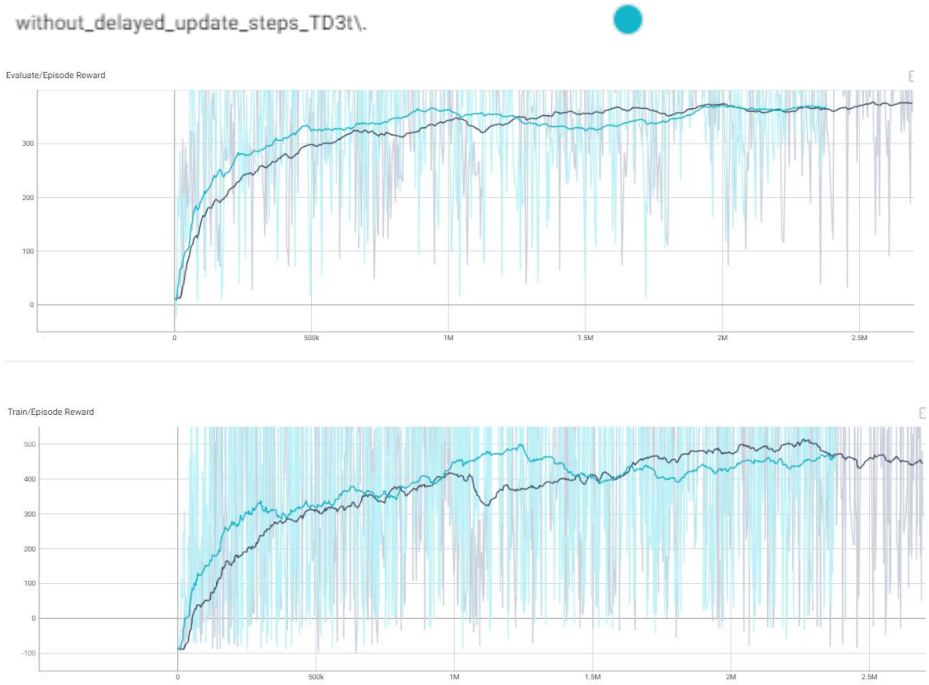





從 Training 圖表中可見 disabling target policy smoothing 的 TD3 的 curve 起伏比 enabling 的 TD3 還要劇烈，所以我們推測 disabling smoothing 的 TD3 對於 noise 更加敏感。

3. Screenshot of Tensorboard training curve and compare the impact of delayed update steps and compare the results, and explain(5%).



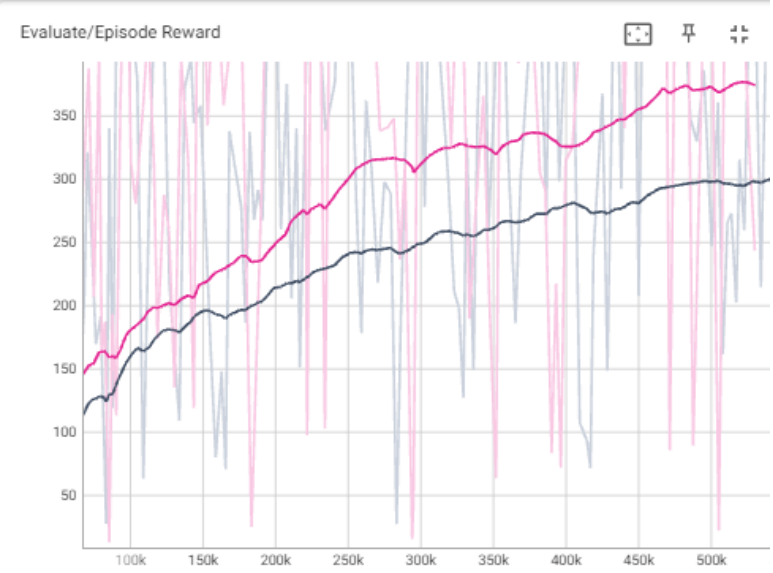


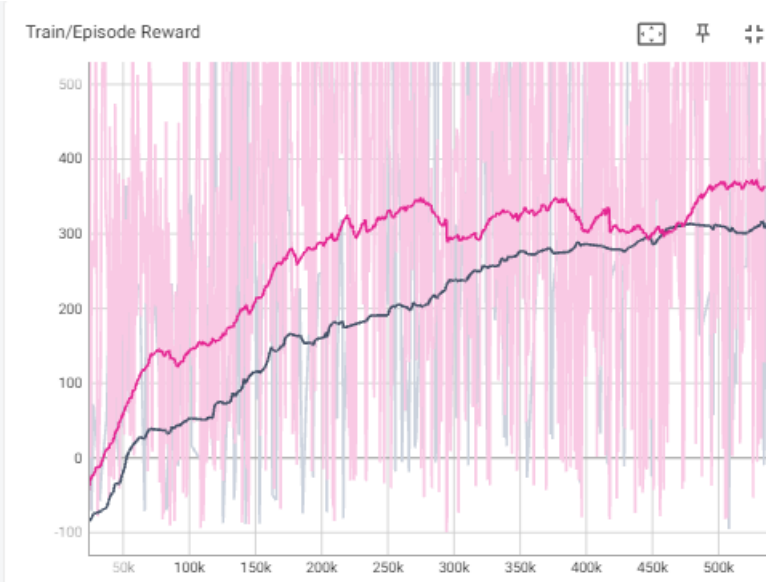

若用 without delayed update steps 的 TD3 來 training，從 Train 的圖表可以看到 reward curve 起伏比 delayed update steps 的 TD3 來的大，推測 without delayed 的 TD3 在 learning 時，對 noise 的抗性比有 delayed 的 TD3 還低，才導致較為不穩定。

而且 without delayed 的 TD3 在 Train 圖表的 reward curve 較早趨於平緩，所以推測 without delayed 的 TD3 會比較早進入收斂。

4. Screenshot of Tensorboard training curve and compare the effects of adding different levels of action noise (exploration noise) in TD3, and explain(5%).






因為 Gaussian Noise 在每個 time step 都是隨機且不相關的；OU Noise 則是會隨著時間而減少，所以我們可以看到兩個圖表中，加入 OU Noise 的 TD3 得到的 reward 較 Gaussian Noise 的 TD3 來的高。
另外，從兩個圖表可見，OU Noise 的 TD3 的 reward curve 較晚變成平緩的狀態，所以推測 OU Noise 的 TD3 較 Gaussian Noise 的 TD3 不易收斂。